

Chapter-6 Electromagnetic Induction.

- 1) Find the self inductance per unit length of a long solenoid of radius R carrying n turns per unit length.

Here,

Let,

l = length of long solenoid

A = cross-sectional area

n = no. of turns per unit length

R = radius R



we know

the flux linkage = $N\phi_B = (nl)(BA)$

B = magnitude of the magnetic field within the solenoid

Thus,

the self inductance of solenoid,

$$\begin{aligned} L &= \frac{N\phi_B}{I} = \frac{nlBA}{I} \\ &= \frac{nl\mu_0 n I A}{I} \\ &= (\mu_0 n^2 A)l \end{aligned}$$

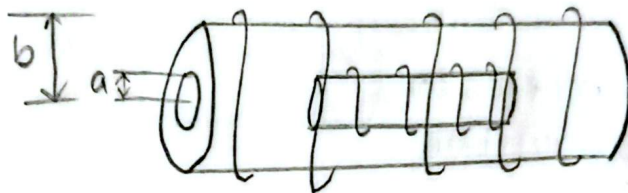
Therefore, the self inductance per unit length of a long solenoid

$$\frac{L}{l} = \mu_0 n^2 A$$

- 2) A short solenoid (length l and radius a , with n_1 turns per unit length) lies on the axis of a very long solenoid (radius b , n_2 turns per unit length). Current I flows in the short solenoid. What is the flux through the long solenoid? What is the mutual induction?

Soln:

Let the current I flows in the long solenoid.



The field inside the long solenoid is constant.

$$B = \mu_0 n_2 I$$

So, the total flux through the short solenoid is

$$\phi_s = (n_1 l) [(\mu_0 n_2 I)(\pi a^2)]$$

According to the reciprocity theorem,

This is also the flux through the long solenoid when the current I flows in the short solenoid.

$$\phi_L = \phi_s = (n_1 l) [(\mu_0 n_2 I)(\pi a^2)]$$

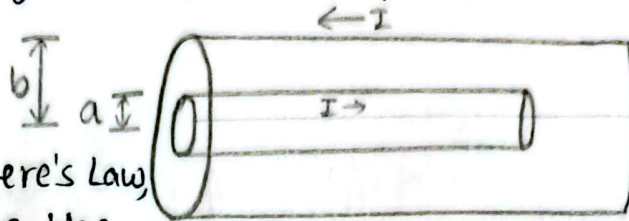
$$= [\mu_0 \pi a^2 n_1 n_2 l] I$$

Mutual Inductance,

$$M = \mu_0 \pi a^2 n_1 n_2 l \quad [\because \phi = M I]$$

3) A long coaxial cable carries current I (the current flows down the surface of the inner cylinder, radius a , and back along the outer cylinder, radius b) as shown in figure. Find the magnetic energy stored in a section of length l . Also find coefficient of self-induction.

Soln:



According to Ampere's Law, the field between the cylinders is

$$B = \frac{\mu_0 I}{2\pi r}$$

Else where, the field is zero.

Thus, the energy per unit volume is

$$U_B = \frac{B^2}{2\mu_0} = \frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi r} \right)^2$$

The energy in a cylindrical shell of length l , radius r , and thickness dr , then is

$$dW = \left[\frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi r} \right)^2 \right] (2\pi r dr l)$$

Therefore, the magnetic energy stored in a section of length l is,

$$W = \frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi} \right)^2 2\pi l \int_a^b \frac{dr}{r}$$

$$\therefore W = \frac{\mu_0 I^2 l}{4\pi} \ln\left(\frac{b}{a}\right) = \frac{1}{2} \left[\frac{\mu_0 l}{2\pi} \ln\left(\frac{b}{a}\right) \right] I^2$$

The self inductance is

$$L = \frac{\mu_0 l}{2\pi} \ln\left(\frac{b}{a}\right) \quad \left[\because W = \frac{1}{2} L I^2 \right]$$

4) Find the energy stored in a section of a length l of a long solenoid (radius R and current I , n turns per unit length) and hence obtain the expression for energy density.

Soln:

The energy stored in a section of length l of a long solenoid is:

$$W = \frac{1}{2\mu_0} \int_{\text{all space}} B^2 d\tau$$

$$= \frac{1}{2\mu_0} B^2 \int d\tau$$

$$= \frac{1}{2\mu_0} \times (\mu_0 n I)^2 \times (\pi R^2 l)$$

$$\therefore W = \frac{1}{2} \mu_0 n^2 \pi R^2 l I^2$$

The energy density is

$$U_B = \frac{B^2}{2\mu_0} = \frac{(\mu_0 n I)^2}{2\mu_0}$$

$$\therefore U_B = \frac{1}{2} \mu_0 n^2 I^2$$